Spacecraft ACDS Homework 2

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1 Safe Mode

1. Done in code.
2. To find the desired angular velocity vector in the body frame corresponding to a spin of 10 RPM about the solar panel vector we multiply the magnitude by the direction of the solar panel normal in the body frame. $^B\omega = ||\omega||^B n$
$^B\omega = \begin{bmatrix} -7.07 \\ 7.07 \\ 0 \end{bmatrix}$ rad/s
3. Using dynamic balance and superspin, we are able to stabilize to spin the satellite about the angular velocity vector in the direction of the solar panels. The no wobble condition is the following

$$\omega \times (J\omega + \rho) = 0 \tag{1}$$

along with the super spin condition $\rho^T\omega = \rho_s\omega_s$ to get a unique solution ($\omega_s = ||\omega||$ and ρ_s is obtained from the superspin condition $\rho_s \geq \omega_s(J_{33} - J_s)$). We choose $J_3 3$ here because it is the minimum state. The term J_s is the inertia in the desired spin axis frame $J_s = (\frac{\omega}{\omega_s})^T J(\frac{\omega}{\omega_s})$.
4. The gyrostat equation is

$$J^B\omega +^B \dot{\rho} +^B \omega \times (J^B\omega + \rho) =^B \tau \tag{2}$$

where J is the inertia in the body frame, $^B\omega$ is the angular velocity in the body frame, ρ is the rotor momentum, and $\dot{\rho}$ is the rotor momentum rate.
5. The perturbation on the initial state has a magnitude of 0.1 $\frac{rad}{s}$ and sampled from a normal distribution. The result of the simulation is stable about the solar panel normal vector as shown in Figure 1.

2 Spacecraft Dynamics

Figure 2 shows the attitude quaternion components for a stable spin about the solar panel normal vector. Figure 3 shows the poining error in degrees from the solar panel normal vector and the sun vector that is assumed to be at [1,0,0] in ECI.

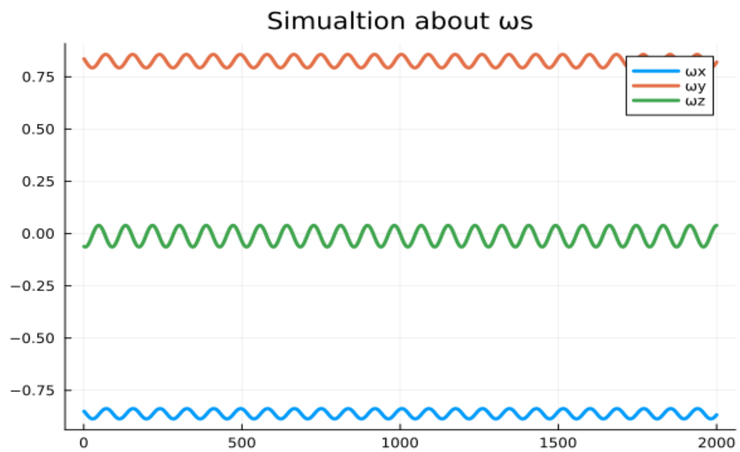

Figure 1: Angular velocity simulation about the solar panel normal vector

3 Attitude Sensors

The attitude sensors onboard the spacecraft are two commercial star trackers. The reported pointing accuracy is ± 0.002 degrees as the 1-σ standard deviation [1]. From this specification we must convert the angular metric into an uncertainty on a bearing vector. We define a bearing measurement at time i as b_i. The covariance is dependent on the direction of the bearing measurement, as the uncertainty will be in the orthogonal subspace to b_i. To do this, we can use the projection $(I - b_i b_i^T)$ and multiply it by our variance converted into radians (this is valid for small angles, which is the case for this sensor). The equation for the covariance matrices is

$$P_s = \sigma^2 (I - b_i b_i^T) \tag{3}$$

We can then sample a noisy measurement b_n by

$$b_n = b_i + \sqrt{P_s} d \tag{4}$$

where d is a 3 dimensional vector sampled from a normal distribution. To verify that the error statistics are consistent, we can take the arc cosine of the dot product between the normal body measurement and the true body measurement. This operation gives us the angle difference between both vectors and most should fall within the 1-sigma range. As seen in Figure 4 the average angle difference is close to the 0.002 deg metric from the star tracker specification sheet.

4 Static Attitude Estimation

For 100 measurements, we ran a MonteCarlo simulation of both the SDP and SVD method of solving Wabha's problem. The rotation error in degrees are

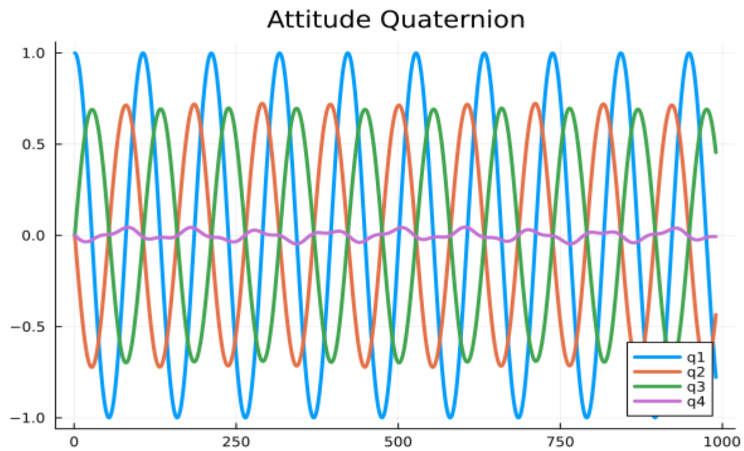

Figure 2: Components of the Attitude Quaternion

very similar and shown in Figures 5 and 6. Using BenchmarkTools.jl in Julia, we are also able to obtain an average time for each appraoch to run. The average time for the SDP method was 0.00184 seconds while the SVD solution ran in 7.007×10^{-6} seconds.

Github Repository: `https://github.com/vegaf1/acds-mit-course`

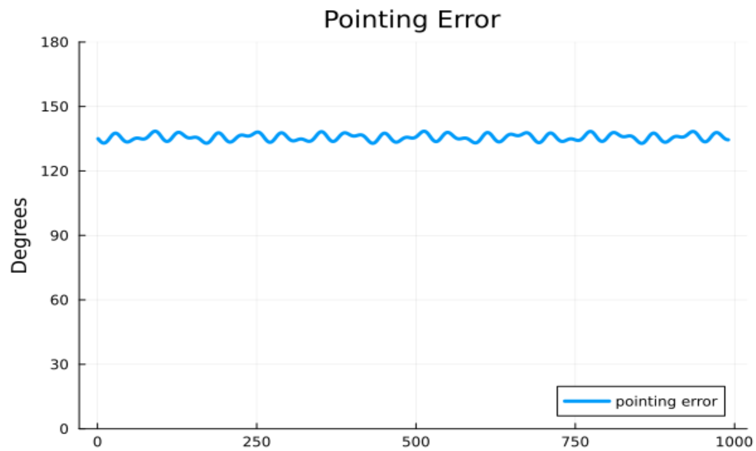

Figure 3: Pointing Error with the sun vector

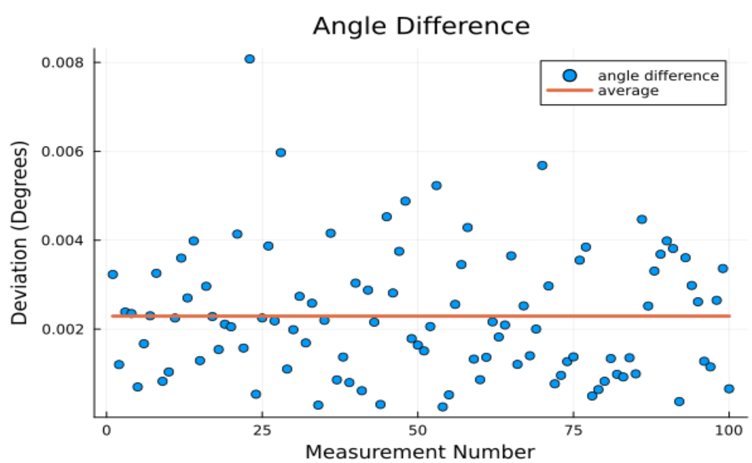

Figure 4: Angle difference between the noisy measurements and true measurements in the body frame

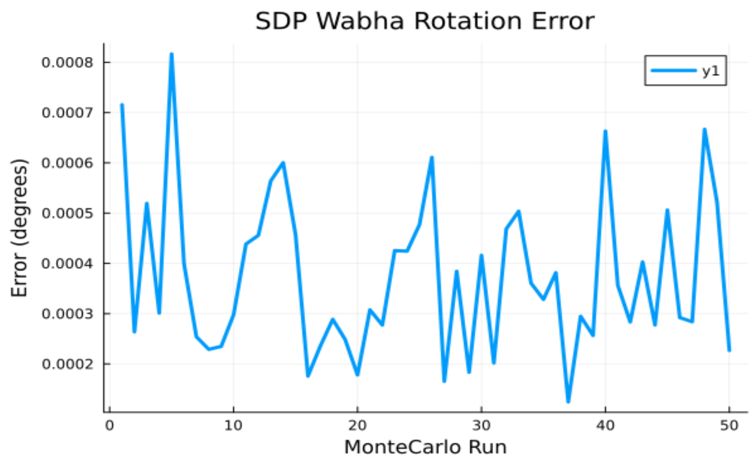

Figure 5: Rotation Error in degrees from solving Wabha's problem using an SDP

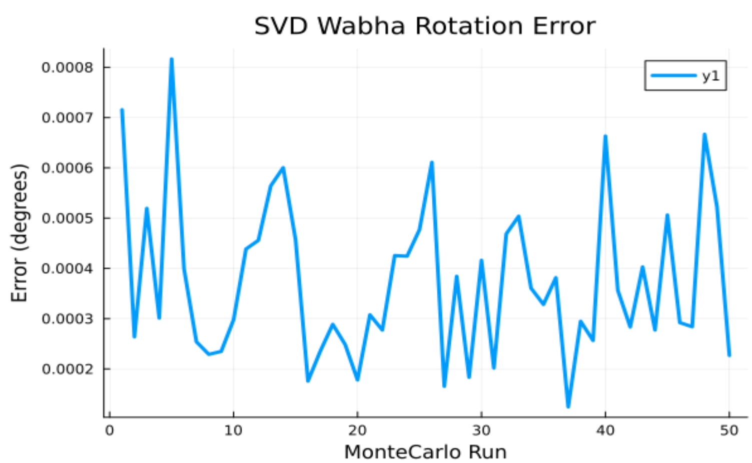

Figure 6: Rotation Error in degrees from solving Wabha's problem using an SVD